

● EASY

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

04

10 answers · Rationale-rich · Teach from doc

Q1**C****C.** 0, 2

Choice C is correct. The vertex of the graph of a quadratic function in the xy -plane is the point at which the graph is either at its minimum or maximum y -value. In the graph shown, the minimum y -value occurs at the point 0, 2.

Choice A is incorrect. The graph shown doesn't pass through the point 0, - 2.

Choice B is incorrect. The graph shown doesn't pass through the point 0, - 3.

Choice D is incorrect. The graph shown doesn't pass through the point 0, 3.

Q2**B****B.** 16

Choice B is correct. The value of $f144$ is the value of $f x$ when $x = 144$. It's given that the function f is defined by $f x = 4 + \sqrt{x}$. Substituting 144 for x in this equation yields $f144 = 4 + \sqrt{144}$. Since the positive square root of 144 is 12, it follows that this equation can be rewritten as $f144 = 4 + 12$, or $f144 = 16$. Therefore, the value of $f144$ is 16.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f1,296$, not $f144$.

Choice D is incorrect. This is the value of $f5,184$, not $f144$.

Q3**C****C.** 23

Choice C is correct. The value of $f(2)$ is the value of $f(x)$ when $x = 2$. Substituting 2 for x in the given function yields $f(2) = 2^3 + 15$, or $f(2) = 8 + 15$, which is equivalent to $f(2) = 23$. Therefore, the value of $f(2)$ is 23.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $f(2)$ when $f(x) = x^3 + 15$, rather than $f(x) = x^3 + 15$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**B****B.** 5, 0

Choice B is correct. An x -intercept of a graph in the xy -plane is a point at which the graph crosses the x -axis. The graph shown crosses the x -axis at the point 5, 0. Therefore, the x -intercept of the graph shown is 5, 0.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5

A

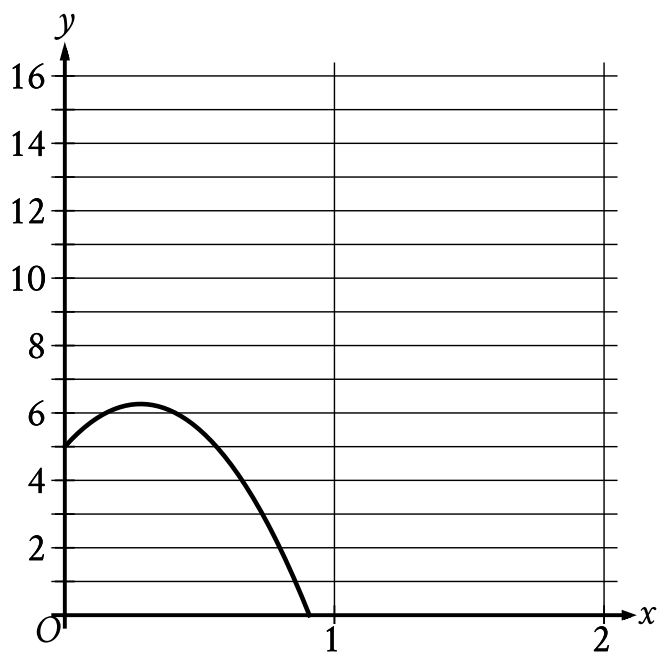
A. The estimated value of the tablet was \$ 225 when it was purchased.

Choice A is correct. It's given that the graph shown gives the estimated value y , in dollars, of a tablet as a function of the number of months since it was purchased, x . The y -intercept of a graph is the point at which the graph intersects the y -axis, or when x is 0. The graph shown intersects the y -axis at the point 0, 225. It follows that 0 months after the tablet was purchased, or when the tablet was purchased, the estimated value of the tablet was 225 dollars. Therefore, the best interpretation of the y -intercept is that the estimated value of the tablet was \$ 225 when it was purchased.

Choice B is incorrect. The estimated value of the tablet 24 months after it was purchased was \$ 50, not \$ 225.

Choice C is incorrect. The estimated value of the tablet had decreased by \$ 225 - \$ 50, or \$ 175, not \$ 225, in the 24 months after it was purchased.

Choice D is incorrect and may result from conceptual errors.

Q6**B****B.**

- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 6.3).
- The parabola passes through the following points:
 - (0 comma 5)
 - approximately (0.3 comma 6.3)
 - approximately (0.6 comma 5)

Choice B is correct. It's given that y represents the height, in feet, of the ball x seconds after it was launched. It's also given that during the first part of an experiment, a ball was launched from a 7-foot-tall platform. Therefore, the y -coordinate of the y -intercept of the given graph, 7, represents the platform height, in feet. During the second part of the experiment, the platform the ball was launched from was 2 feet shorter than the platform in the first part of the experiment. It follows that the height of the platform in the second part of the experiment was $7 - 2$ feet, or 5 feet. Therefore, the y -coordinate of the y -intercept of the graph representing the second part of the experiment must be 5. Only choice B satisfies this condition.

Choice A is incorrect. This could represent the graph if the ball were launched from a platform that was about 3 feet shorter rather than 2 feet shorter.

Choice C is incorrect. This could represent the graph if the ball were launched from a platform that was 2 feet taller rather than 2 feet shorter.

Choice D is incorrect. This could represent the graph if the ball were launched from a platform that was twice as tall rather than 2 feet shorter.

Q7

D

D. $\frac{1}{18}$

Choice D is correct. It's given that $fx = \frac{1}{6x}$. Substituting 3 for x in this equation yields $f3 = \frac{1}{63}$, or $f3 = \frac{1}{18}$. Therefore, when $x = 3$, the value of fx is $\frac{1}{18}$.

Choice A is incorrect. This is the value of fx when $x = 0.5$.

Choice B is incorrect. This is the value of fx when $x = 1$.

Choice C is incorrect. This is the value of fx when $x = 1.5$.

Q8

3

The correct answer is 3. For the graph of the exponential function f shown, where $y = fx$, it's given that the y -intercept of the graph is 0, y . The graph intersects the y -axis at the point 0, 3. Therefore, the value of y is 3.

Q9

.5

Accept also: $\frac{1}{2}$

The correct answer is $\frac{1}{2}$. The value of $h2$ is the value of hx when $x = 2$. Substituting 2 for x in the given equation yields $h2 = \frac{8}{52+6}$, which is equivalent to $h2 = \frac{8}{16}$, or $h2 = \frac{1}{2}$. Therefore, the value of $h2$ is $\frac{1}{2}$. Note that $\frac{1}{2}$ and .5 are examples of ways to enter a correct answer.

Q10

B

B. $h(n) = 22(0.85)^n$

Choice B is correct. It's given that for the function h , hn is the estimated maximum height, in feet, of the ball after it has hit the ground n times. It's also given that the function h estimates that the maximum height reached after each time the ball hits the ground is 85% of the maximum height reached after the previous time the ball hit the ground. It follows that h is a decreasing exponential function that can be written in the form $hn = a\frac{p}{100}^n$, where a is the initial height, in feet, the ball was dropped from and the function estimates that the maximum height reached after each time the ball hits the ground is $p\%$ of the maximum height reached after the previous time the ball hit the ground. It's given that the ball is dropped from an initial height of 22 feet. Therefore, $a = 22$. Since the function h estimates that the maximum height reached after each time the ball hits the ground is 85% of the maximum height reached after the previous time the ball hit the ground, $p = 85$. Substituting 22 for a and 85 for p in the equation $hn = a\frac{p}{100}^n$ yields $hn = 22\frac{85}{100}^n$, or $hn = 22(0.85)^n$.

Choice A is incorrect. This function estimates that the maximum height reached after each time the ball hits the ground is 22%, not 85%, of the maximum height reached after the previous time the ball hit the ground.

Choice C is incorrect. This function estimates that the ball is dropped from an initial height of 85 feet, not 22 feet, and that the maximum height reached after each time the ball hits the ground is 22%, not 85%, of the maximum height reached after the previous time the ball hit the ground.

Choice D is incorrect. This function estimates that the ball is dropped from an initial height of 85 feet, not 22 feet.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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